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THIRD TECHNICAL PANEL REVIEW REPORT

**INDIGENOUS DEVELOPMENT
OF DNA TAGGING SYSTEM FOR
INDUSTRIES**

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Islamabad, Pakistan — August 2026

Signatures and Institutional Approvals

This formal project review report delineates the comprehensive technical, experimental, and empirical milestones accomplished under the defense and industrial-sponsored project “**Indigenous Development of DNA Tagging System for Industries**” at the National University of Sciences and Technology (NUST).

The signatures below verify that the research methodologies, algorithmic bioinformatic pipelines, nanomaterial encapsulation processes, wet-lab qPCR multiplex validations, and digital infrastructure documented herein have been executed and verified in compliance with national forensic standards and NUST institutional guidelines.

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Executive Summary & Project Mandate

1.1 Executive Summary

The overarching objective of the **DNAx Project** (*Indigenous Development of DNA Tagging System for Industries*) is to design, synthesize, encapsulate, and commercially deploy an unforgeable, forensic-grade biological taggant ecosystem. Initiated on **January 2, 2026**, under the joint auspices of the Atta-ur-Rahman School of Applied Biosciences (ASAB) and the School of Chemical and Materials Engineering (SCME) at NUST, this project bridges advanced computational synthetic biology, inorganic nanotechnology, automated industrial manufacturing, and digital chain-of-custody tracking.

Counterfeit goods, illicit supply chain diversion, and untraceable ordnance in high-stakes forensic investigations represent critical economic and national security threats. Conventional optical markers (such as standard barcodes, QR stickers, and fluorescent dyes) are susceptible to replication, physical ablation, and chemical tampering. By harnessing the astronomical information density and biochemical specificity of synthetic DNA, the DNaX ecosystem provides a covert, tamper-proof, and legally unassailable authentication standard.

Over the reporting period culminating in the **Third Technical Panel Review (25th August 2026)**, the research consortium has transitioned definitively from theoretical modeling to full physical validation. Key technical breakthroughs achieved to date include:

1. **Complete In-House Bioinformatic Suite (DNAX Signet[©]):** Developed and deployed a robust pipeline for physical sizing, de novo linear DNA sequence synthesis, thermodynamic primer/probe optimization ($T_m \geq 68.0^\circ\text{C}$), and local BLAST genome screening.
2. **Wet-Lab DNA Synthesis, PCR Up-Scaling & Formulation:** Successfully

synthesized a target 200 bp synthetic DNA construct (Lot No. BJ033535) and established high-yield PCR amplification protocols to formulate 1.0 Liter of concentrated industrial taggant solution.

3. **Silica Core-Shell Encapsulation & Characterization:** Successfully synthesized SiO_2 spherical nanoparticles (200–300 nm), achieved cationic charge reversal using CTAB surfactant (δ^+), encapsulated anionic DNA (δ^-), and demonstrated thermal resilience exceeding 800°C with rare-earth luminescent phosphors ($\text{SiO}_2@\text{SnO}_2 : \text{Eu}^{3+}$).
4. **Industrial Substrate Integration:** Demonstrated practical application across commercial pharmaceutical packaging, homeopathic bottles, excise tax stamps/tobacco items, and designed an automated roll-to-roll spray/curing assembly line.
5. **Wet-Lab qPCR Multiplex Proof-of-Concept:** Conducted real-time quantitative PCR (qPCR) validation using a 4-channel fluorophore probe library (**FAM, HEX, Texas Red/ROX, Cy5**), achieving unambiguous, simultaneous detection across all 4 optical channels with zero cross-talk.
6. **Unified 4-Tier Digital Architecture:** Engineered the interconnected software ecosystem comprising DNAx Signet (Research), DNAx Manufacturing (Production), DNAx Vendor (Supply Chain), and DNAx Forensics (Legal/Agency Verification) backed by central role-isolated database vaults.

1.2 Project Deliverables and Scope of Work

The project scope is structured around six core technical deliverables, as officially mandated in the institutional project agreement:

[Figure 1.1: Schematic Representation of the 6 Technical Pillars
of the DNAx Ecosystem]

Figure 1.1: The six core technical deliverables of the Indigenous DNA Tagging System.

1. **Deliverable 1: DNA Taggant Design by DNAx Signet[®] Software:** Algorithmic sequence generation, thermodynamic stability guarantees, and in silico orthogonality screening against environmental genomes.

2. **Deliverable 2: DNA Taggant Synthesis & Up-Scaling:** Wet-lab biochemical synthesis, plasmid cloning, PCR bulk amplification, and formulation into carrier buffers.
3. **Deliverable 3: DNA Taggant Coating & Nanomaterial Encapsulation:** Core-shell inorganic encapsulation in functionalized silica and phosphor matrices for extreme environmental survival.
4. **Deliverable 4: DNA Taggant Integration into Products:** Physical dispersion and bonding into consumer packaging, high-value liquid pharmaceuticals, and metallic or security materials.
5. **Deliverable 5: DNA Taggant Extraction, Detection by qPCR & Identification by DNAx Forensics[®]:** Chemical recovery protocols, real-time multiplex qPCR assay execution, and algorithmic barcode deconvolution for agency verification.
6. **Deliverable 6: DNA Taggant Digital Management System (DMS):** Cloud/local dual database synchronization, blind laboratory sample tracking, and immutable multi-party transaction logging.

1.3 Milestone Progress and Work-Done Trajectory

Figure 1.2 and Table 1.1 summarize the cumulative progress across the six deliverables as presented to the Technical Panel on August 25, 2026.

[Figure 1.2: Cumulative Work Done Progress Bar Graph across 6 Technical Pillars]

Figure 1.2: Progress matrix depicting percentage completion across the project deliverables.

Table 1.1: Deliverable Status and Empirical Achievement Summary (August 2026)

No.	Technical Deliverable	Progress (%)	Key Empirical Milestones Completed
1	DNA Taggant Design (DNAx Signet)	100%	Full 6-stage algorithmic synthesis engine, BLAST safety screen, nearest-neighbor T_m calculation.
2	DNA Taggant Synthesis & Up-Scaling	90%	200 bp construct synthesized (Lot BJ033535), gel electrophoresis confirmed, 1L taggant buffer prepared.
3	DNA Taggant Nanocoating	80%	Stöber SiO ₂ core synthesis, CTAB functionalization, SEM, EDX, XRD, FTIR, and PSA laser diffraction completed.
4	Product Integration	75%	Tested on pharmaceutical bottles, excise stamps, and engineered roll-to-roll automated spray hardware.
5	DNA Recovery & qPCR Detection	85%	Extraction protocols optimized, 4-probe multiplex qPCR executed with 100% fluorophore channel detection.
6	Digital Management System (DMS)	85%	Developed DNAx Signet, Manufacturing, Vendor, and Forensics apps with centralized SQL/Firestore ledger.

DNA Taggant Design, Combinatorics & Software Engine

2.1 Introduction to DNA-Based Information Storage

Deoxyribonucleic acid (DNA) offers unprecedented theoretical data storage density (4.55×10^{18} bytes/gram) and biochemical stability spanning millennia under appropriate physical shielding. In the DNAX ecosystem, synthetic DNA serves as an ultra-high-entropy physical security identifier. Rather than encoding raw text, the sequence is engineered to function as a multi-stage molecular signature, containing:

1. **Conserved Flanking Primer Binding Sites:** Dedicated 20 bp forward and reverse primer sequences that enable targeted polymerase chain reaction (PCR) amplification from complex environmental matrices.
2. **Internal Multiplex Optical Probe Channels:** Contiguous 24 bp target hybridization sequences recognized by fluorescently labeled dual-quenched TaqMan hydrolysis probes during real-time qPCR.
3. **Randomized Orthogonal Spacer Segments:** In silico screened nucleotide blocks that prevent steric clash, eliminate secondary stem-loop structures, and guarantee zero homology with environmental or human genomes.

2.2 The 6-Stage Bioinformatic Design Pipeline (DNAX Signet[©])

The proprietary in-house bioinformatic platform, **DNAX Signet[©]**, guides researchers through a strict 6-stage algorithmic synthesis pipeline (Figure 2.1):

**[Figure 2.1: Architectural Flowchart of the 6-Stage DNAX Signet
Bioinformatic Pipeline]**

Figure 2.1: The 6-stage bioinformatic assay pipeline implemented in DNAX Signet[©].

1. **Stage 1: Physical Sizing:** The user defines the target construct length (standard: 200 bp, expandable up to 1000 bp). The system dynamically calculates the physical constraints:

$$L_{\min} = L_{\text{Fwd}} + L_{\text{Rev}} + \sum_{i=1}^{N_{\text{probes}}} L_{\text{probe},i} + (N_{\text{probes}} + 1) \cdot L_{\text{spacer}} \quad (2.1)$$

where $L_{\text{Fwd}} = 20$ bp, $L_{\text{Rev}} = 20$ bp, $L_{\text{probe}} = 24$ bp, and $L_{\text{spacer}} \geq 6$ bp.

2. **Stage 2: De Novo DNA Sequence Synthesis:** An orthogonal sequence is algorithmically generated using a constrained Markov random generator with GC-windowing ($48.0\% \leq \text{GC} \leq 52.0\%$), zero homopolymer runs (≥ 4 identical consecutive nucleotides), and complete exclusion of major restriction enzyme recognition sites (e.g., *EcoRI*, *BamHI*, *HindIII*, *BsaI*).
3. **Stage 3: In Silico Homology & Local BLAST Screening:** The sequence is screened locally against the NCBI non-redundant nucleotide database and an internal matrix of all previously registered DNAX tags. Sequences exhibiting $\geq 75\%$ identity over any 20 bp window with pathogenic or common environmental microorganisms are automatically rejected to eliminate false positives and biosafety compliance issues.
4. **Stage 4: PCR Primer Pair Generation:** Orthogonal forward and reverse primers are designed using SantaLucia nearest-neighbor thermodynamic parameters:

$$T_m = \frac{\Delta H^\circ}{\Delta S^\circ + R \ln(C_T/4)} + 16.6 \log_{10}[\text{Na}^+] \quad (2.2)$$
 Primers are constrained to $L = 20$ bp, $T_m = 58.5 \pm 1.0^\circ\text{C}$, $\Delta T_m \leq 0.5^\circ\text{C}$, and $\Delta G_{3'\text{-dimer}} > -2.0$ kcal/mol.
5. **Stage 5: Multiplex qPCR Hydrolysis Probes:** The engine assigns up to 8 distinct hybridization target zones within the construct core. Probes are strictly standardized to 24 bp and $T_m \geq 68.0^\circ\text{C}$ (strictly $+10.0^\circ\text{C}$ above primers to ensure

complete probe binding before primer extension). Probes are programmatically forbidden from having a 5'-Guanine to prevent instantaneous fluorescence quenching.

6. Stage 6: Review, Cryptographic QR Generation & Database Vaulting:

The verified construct is signed with an immutable SHA-256 hash token (e.g., DNAX-QR-8F39A1C4), logged to the central database, and exported as a formal molecular specification dossier.

2.3 Probe Chain Architecture and Molecular Anatomy

The structural anatomy of the linear double-stranded DNA (dsDNA) construct is illustrated in Figure 2.2. Four contiguous probe segments (**Probe 1 FAM**, **Probe 2 HEX**, **Probe 3 Texas Red/ROX**, **Probe 4 Cy5**) form the information payload, flanked on both ends by the forward and reverse primer annealing zones.

[Figure 2.2: Molecular Anatomy of the 200 bp Taggant:
4-Component Linked Probe Chain with Flanking Primers]

Figure 2.2: Molecular architecture of the synthesized DNA taggant: (A) The 4 component probes, (B) Linked probe chain assembly, and (C) Primer annealing to form the completed double-stranded tag.

2.4 Mathematical Combinatorics of the 200 bp Sequence Space

The diversity and unforgeability of the DNAx system stem from the vast combinatorial sequence space available in biological polymers.

2.4.1 Raw Sequence Combinations

For a linear DNA construct of length $N = 200$ base pairs, with 4 possible nucleotide choices (A, T, C, G) per position, the total possible unique sequence combinations S_{total} is given by:

$$S_{\text{total}} = 4^{200} = (2^2)^{200} = 2^{400} \approx 2.58 \times 10^{120} \quad (2.3)$$

This number (2.58×10^{120}) exceeds the total estimated number of atoms in the observable universe ($\approx 10^{80}$) by 40 orders of magnitude, rendering brute-force guessing or accidental environmental duplication physically impossible.

2.4.2 Modular Building Block Library Combinations

To enable rapid modular synthesis and cost-effective multiplex qPCR detection, DNAx employs a combinatorial building block strategy. In this architecture, constructs are assembled from a standardized library of pre-validated, orthogonal 24 bp probe blocks.

For a library of $M = 72$ unique, non-interfering DNA building blocks, choosing $k = 4$ distinct blocks per taggant (without replacement, where order is tracked or unordered multiplexing is applied) yields:

$$C(M, k) = \binom{72}{4} = \frac{72!}{4!(72-4)!} = \frac{72 \times 71 \times 70 \times 69}{4 \times 3 \times 2 \times 1} = \frac{24,690,960}{24} = 1,028,790 \quad (2.4)$$

Thus, a compact physical library of only 72 synthesized probe blocks yields **over one million** ($> 1.028 \times 10^6$) **mathematically unique, fully orthogonal industrial taggants**.

Physical DNA Synthesis, PCR Amplification & Bulk Formulation

3.1 Introduction and Chemical Synthesis

Following the in silico design and thermodynamic verification in DNAX Signet[©], physical synthesis of the validated 200 bp sequence was initiated. The complete double-stranded DNA construct was commercially cloned into a high-copy cloning vector (**pUC57-Amp**) under **Lot No. BJ033535** and purified via standard silica spin-column miniprep.

In parallel, high-performance liquid chromatography (HPLC)-purified single-stranded DNA primers and dual-labeled fluorophore/quencher TaqMan probes were acquired for amplification and validation assays:

- **Forward Primer:** 20 nt ($T_m \approx 59.2^\circ\text{C}$, GC = 50.0%).
- **Reverse Primer:** 20 nt ($T_m \approx 58.8^\circ\text{C}$, GC = 50.0%).
- **Probe A (FAM Channel):** 24 nt, 5'-FAM reporter, 3'-BHQ-1 dark quencher ($T_m = 69.5^\circ\text{C}$).
- **Probe B (HEX Channel):** 24 nt, 5'-HEX reporter, 3'-BHQ-1 dark quencher ($T_m = 70.1^\circ\text{C}$).
- **Probe C (Texas Red / ROX Channel):** 24 nt, 5'-Texas Red reporter, 3'-BHQ-2 dark quencher ($T_m = 69.8^\circ\text{C}$).
- **Probe D (Cy5 Channel):** 24 nt, 5'-Cy5 reporter, 3'-BHQ-3 dark quencher ($T_m = 70.4^\circ\text{C}$).

[Figure 3.1: Physical Reagent Vials of Synthesized Primers, 4 Probes, and Miniprep Taggant (Lot BJ033535)]

Figure 3.1: Synthesized physical inventory: Forward/Reverse Primers, Probes A–D, and the 200 bp pUC57 plasmid miniprep stock.

3.2 Polymerase Chain Reaction (PCR) Amplification Protocol

To generate high-concentration linear dsDNA fragments for industrial formulation, bulk PCR amplification was established using a high-fidelity DNA polymerase master mix.

3.2.1 Thermal Cycling Program

The optimized thermal cycling conditions for bulk amplification of the 200 bp taggant are detailed in Table 3.1.

Table 3.1: Optimized PCR Thermal Cycling Parameters for 200 bp Taggant Amplification

Reaction Step	Temperature (°C)	Hold Time	Number of Cycles
Initial Denaturation	95°C	5 min	1
Denaturation	95°C	30 sec	35 cycles
Annealing	55°C	30 sec	
Extension	72°C	45 sec	
Final Extension	72°C	7 min	1
Hold	4°C	∞	1

3.2.2 Reaction Composition

Reactions were prepared in 25.0 μL total volume as specified in Table 3.2. Two reaction variations (Reaction 1 with 1.0 μL template vs. Reaction 2 with 2.0 μL template) were evaluated to assess yield and template saturation.

Table 3.2: PCR Master Mix Recipe (25.0 μL Total Reaction Volume)

Reagent Component	Reaction 1 Volume (μL)	Reaction 2 Volume (μL)
10X Standard Taq Reaction Buffer	2.5 μL	2.5 μL
MgCl ₂ (25 mM stock)	1.5 μL	1.5 μL
Forward Primer (10 μM)	1.0 μL	1.0 μL
Reverse Primer (10 μM)	1.0 μL	1.0 μL
Taq DNA Polymerase (5 U/ μL)	1.0 μL	1.0 μL
Template DNA (pUC57-Taggant Miniprep)	1.0 μL	2.0 μL
Nuclease-Free Distilled Water (ddH ₂ O)	16.0 μL	15.0 μL
Total Reaction Volume	25.0 μL	25.0 μL

3.3 Agarose Gel Electrophoresis Verification

Amplified PCR products were resolved on a 2.0% (w/v) high-resolution agarose gel prepared in $1 \times$ TAE buffer (40 mM Tris-acetate, 1 mM EDTA, pH 8.3) pre-stained with ethidium bromide (0.5 $\mu\text{g}/\text{mL}$). A 100 bp DNA molecular weight ladder was electrophoresed in parallel at 90 V for 45 minutes and visualized under ultraviolet transillumination (302 nm).

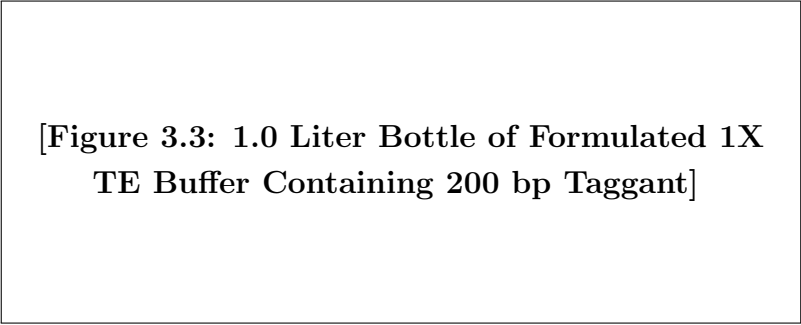
[Figure 3.2: Agarose Gel Electrophoresis Analysis Revealing Intense, Pure 200 bp Amplicon Band]

Figure 3.2: Agarose gel electrophoresis confirming robust, specific amplification of the 200 bp synthetic taggant with zero non-specific banding or primer-dimer artifact formation.

As shown in Figure 3.2, the PCR amplification yielded an intense, sharp band migrating precisely at the expected **200 bp** position. No non-specific high-molecular-weight amplicons or lower primer-dimer bands were detected, confirming complete sequence specificity and optimal primer annealing thermodynamics.

3.4 Bulk Formulation: Preparation of 1.0 Liter Industrial Taggant Solution

To transition from bench-scale molecular biology to scalable industrial application, an optimized carrier buffer protocol was formulated to prepare 1.0 Liter of stable, ready-to-use liquid taggant (Figure 3.3):



[Figure 3.3: 1.0 Liter Bottle of Formulated 1X TE Buffer Containing 200 bp Taggant]

Figure 3.3: Bulk 1.0 Liter preparation of synthetic DNA taggant dissolved in stabilized $1 \times$ TE buffer for commercial industrial integration.

3.4.1 Formulation Protocol for 1 Liter of Taggant Buffer

1. **10.0 mL of 1.0 M Tris-HCl (pH 8.0):** Acts as a primary biological pH buffer, maintaining ionic stability in the range of pH 7.8–8.2.
2. **2.0 mL of 0.5 M EDTA (pH 8.0):** Serves as a potent divalent cation chelator (Mg^{2+} , Ca^{2+}), effectively inactivating trace environmental deoxyribonucleases (DNases) that could otherwise catalyze enzymatic degradation of the taggant.
3. **988.0 mL of Ultra-Pure Sterile Distilled Water (ddH_2O).**
4. **100.0 μL of Concentrated PCR Amplicon Product** (derived from 4–5 bulk PCR tubes, yielding an estimated working copy number of $\sim 10^7$ – 10^8 copies/mL).

The resulting formulation provides an optically transparent, stable aqueous taggant that can be directly metered into printing inks, adhesives, industrial coating suspensions, and liquid pharmaceutical lots without altering the physicochemical properties of the host product.

Nanomaterial Encapsulation, Surface Chemistry & Multi-Modal Characterization

4.1 Nanotechnology Taggant Architecture

While bare synthetic DNA is stable in buffered liquid solutions, harsh industrial environments — such as high-temperature pyrotechnics, metallic projectile casting, explosive detonations, intense UV solar irradiation, and corrosive chemical exposures — demand robust physical protection. To overcome these constraints, the materials science team at SCME-NUST engineered a multi-layered, inorganic core-shell nanoparticle system (Figure 4.1).

[Figure 4.1: Multi-Layer Core-Shell Architecture: Silica Core, Cationic CTAB Bilayer, Encapsulated DNA, Phosphor Dopants, and Outer Ceramic Shield]

Figure 4.1: Schematic cross-section of the concentric DNA taggant composite nanoparticle.

The composite taggant particle comprises four concentric functional layers:

1. **Monodisperse Silica Core (SiO_2 , 200–300 nm):** Serves as an isotropic, opti-

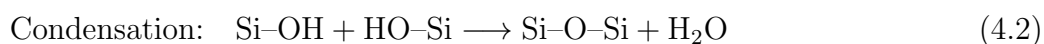
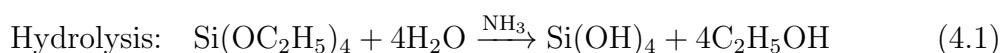
cally transparent, and chemically inert structural scaffold synthesized via the Stöber sol-gel process.

2. **Cationic Surfactant Bilayer (CTAB):** Reverses the native negative surface charge (δ^-) of silica to an electropositive surface (δ^+), driving spontaneous electrostatic condensation of the polyanionic DNA phosphate backbone.
3. **Inorganic Encapsulating Shell (SiO_2):** Formed by secondary base-catalyzed hydrolysis of tetraethyl orthosilicate (TEOS), permanently entrapping the adsorbed DNA strands within a dense, glassy silica matrix.
4. **Luminescent Phosphor Coating & Outer Ceramic Barrier:** Incorporates rare-earth optical dopants ($\text{SnO}_2 : \text{Eu}^{3+}$ and long-afterglow $\text{SrAl}_2\text{O}_4 : \text{Eu}^{2+}, \text{Dy}^{3+}$) to enable instant UV optical screening, sealed within a refractory ceramic shell withstanding temperatures exceeding 800°C .

4.2 Chemical Synthesis Methodology

4.2.1 Stage I: Stöber Sol–Gel Synthesis of Monodisperse SiO_2 Nanoparticles

Spherical silica cores were synthesized using the base-catalyzed Stöber method. Absolute ethanol (99.9%, 30.0 mL), aqueous ammonia solution (25.0% NH_4OH , 3.0 mL), and deionized water (2.0 mL) were homogenized under magnetic stirring at 400 rpm for 10 minutes at ambient temperature (25°C). Tetraethyl orthosilicate (TEOS, 1.5 mL) was introduced in a controlled, dropwise manner over 36 minutes (one drop every 15 seconds). The sol was stirred for 4 hours to ensure quantitative hydrolysis and condensation:



The synthesized nanoparticles were collected by high-speed centrifugation (8,000 rpm, 5 minutes), washed twice in deionized water and twice in ethanol, yielding 0.072 g of pure white SiO_2 nanopowder.

4.2.2 Stage II: CTAB Surface Functionalization and Charge Inversion

Native silica nanoparticles possess a negative zeta potential due to surface deprotonated silanol groups ($\equiv \text{Si}-\text{O}^-$). Because DNA is also polyanionic due to its phosphate backbone ($-\text{PO}_4^-$), direct adsorption is prevented by electrostatic repulsion.

**[Figure 4.2: CTAB Micellar Bilayer Assembly around
Silica Core Facilitating Cationic Charge Reversal]**

Figure 4.2: CTAB self-assembly on silica core providing the cationic ammonium ($N^+(CH_3)_3$) interface for DNA binding.

To induce charge reversal, the washed SiO_2 pellet was dispersed in 30.0 mL ethanol by probe ultrasonication (10 minutes). Cetyltrimethylammonium bromide (CTAB, $C_{19}H_{42}BrN$) was dissolved in ethanol and added dropwise under continuous stirring at 500 rpm. The hydrophobic 16-carbon alkyl tails of CTAB pack into a dense bilayer while the positively charged quaternary ammonium head groups (δ^+) orient outward, creating an electropositive interface (Figure 4.2).

4.2.3 Stage III: DNA Encapsulation within Secondary Silica Shell

The CTAB-functionalized SiO_2 suspension was introduced into the buffered synthetic DNA solution at 500 rpm, facilitating rapid electrostatic adsorption of the DNA constructs onto the positively charged particle surfaces. Ammonia solution was added to adjust the pH to ~ 9.0 , activating secondary sol-gel polymerization.

A secondary volume of TEOS (1.0 mL) was added dropwise over 16 minutes, with an additional 0.6 mL added after 24 hours. The encapsulation reaction proceeded for 2.5 days to allow full growth of the outer amorphous silica network, transforming the suspension into a characteristic high-viscosity protective gel.

4.3 Experimental Characterization Results

The synthesized nanomaterials were subjected to comprehensive structural, chemical, and optical characterization across four complementary modalities: Scanning Electron Microscopy (SEM), Particle Size Analysis (PSA), Fourier Transform Infrared Spectroscopy (FTIR), and X-Ray Diffraction (XRD) (Figure 4.3).

[Figure 4.3: Multi-Modal Characterization Quad-Chart: (A) XRD Phase Profiles, (B) PSA Size Distribution, (C) FTIR Spectra, and (D) High-Resolution SEM Micrographs]

Figure 4.3: Comprehensive experimental characterization quad-chart of the encapsulated DNA taggants.

4.3.1 Scanning Electron Microscopy (SEM) and Energy-Dispersive X-Ray Spectroscopy (EDX)

High-resolution field-emission SEM micrographs (Figure 4.3D) obtained at accelerating voltages of 10–20 kV (11,000 $\times$ and 50,000 $\times$ magnification) revealed highly uniform, spherical nanoparticles with primary diameters tightly distributed between **200 nm and 300 nm**. EDX spectra confirmed stoichiometric purity, displaying dominant silicon ($K_{\alpha} = 1.74$ keV) and oxygen ($K_{\alpha} = 0.52$ keV) peaks with zero transition metal contamination.

4.3.2 Particle Size Analysis (PSA) by Laser Diffraction

Laser diffraction particle size analysis (HORIBA LA-920) of the synthesized SiO₂ cores revealed a monodisperse primary distribution centered at $D_{50} = 3.658 \mu\text{m}$ (mean: $3.807 \mu\text{m}$, representing mild controlled agglomeration in suspension). Following DNA adsorption and secondary silica shell growth, a marked rightward hydrodynamic shift toward the **5–10 μm** region was observed (Figure 4.3B), providing definitive physical proof of secondary shell deposition encapsulating the DNA-silica complexes.

4.3.3 Fourier Transform Infrared Spectroscopy (FTIR)

FTIR spectra (Figure 4.3C) recorded from 4000 cm^{-1} to 400 cm^{-1} conclusively tracked the chemical functionalization at each step:

- **Pure SiO₂:** Exhibited fundamental Si–O–Si asymmetric stretching at 1080 cm^{-1} , Si–O bending at 465 cm^{-1} , and broad surface silanol O–H stretching at 3400 cm^{-1} .

- **CTAB Functionalization:** Introduced characteristic aliphatic C–H stretching doublets at 2920 cm^{-1} ($\nu_{\text{as}}(\text{CH}_2)$) and 2850 cm^{-1} ($\nu_{\text{s}}(\text{CH}_2)$), confirming surfactant grafting.
- **Post-Encapsulated DNA:** Displayed signature DNA phosphate backbone P=O asymmetric stretching at $1220\text{--}1240\text{ cm}^{-1}$, nucleobase C=O vibrations at $1600\text{--}1700\text{ cm}^{-1}$, and deoxyribose C–O stretching at $1000\text{--}1100\text{ cm}^{-1}$, directly confirming intact chemical preservation of the encapsulated genetic material.

4.3.4 X-Ray Diffraction (XRD) Phase Analysis

XRD diffractograms (Figure 4.3A) exhibited a classic broad amorphous halo centered at $2\theta \approx 15^\circ\text{--}30^\circ$, confirming the non-crystalline, isotropic network of the Stöber silica matrix. The lack of sharp crystalline peaks confirms that the secondary silica shell grew without localized phase separation, preserving an amorphous cage structure that shields the encapsulated DNA against thermal and mechanical shear.

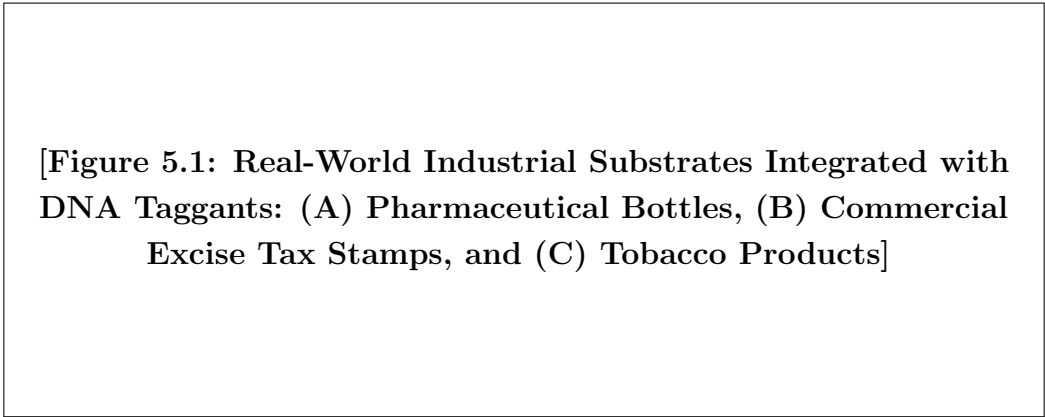
Industrial Product Integration & Automated Roll-to-Roll Spray Machinery

5.1 Introduction to Industrial Taggant Dispersion

A critical requirement of an industrial anti-counterfeiting and forensic tracing standard is seamless, non-destructive integration into commercial manufacturing lines. The DNAX taggant formulation was engineered to be completely invisible, non-toxic, odorless, and chemically inert, ensuring that product aesthetics, physical performance, and regulatory compliance remain entirely uncompromised.

5.2 Product Classes and Substrates Evaluated

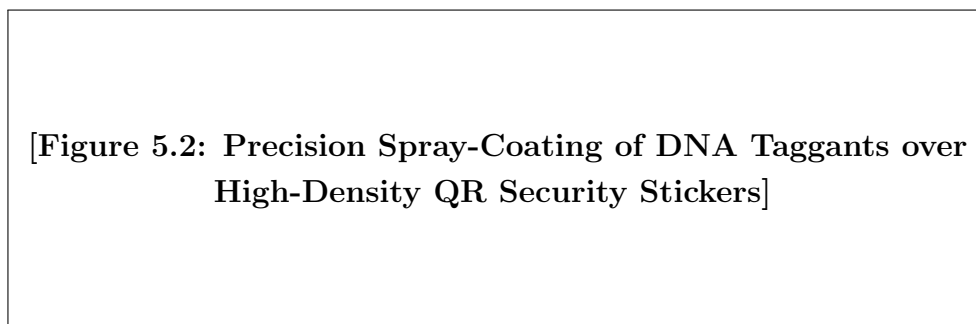
During the reporting period, empirical integration trials were conducted across diverse commercial, medical, and security substrates (Figure 5.1):



[Figure 5.1: Real-World Industrial Substrates Integrated with DNA Taggants: (A) Pharmaceutical Bottles, (B) Commercial Excise Tax Stamps, and (C) Tobacco Products]

Figure 5.1: Photographs of physical commercial products tagged with formulated synthetic DNA constructs.

1. **Pharmaceutical Bottles and Medical Liquid Formulations:** Liquid syrups, dropper bottles (e.g., *Dilan*, *Gelsemium 30*, *Surkhana*), and sealed blister packs were tagged by micro-dosing taggant solution into the packaging varnish and cap seal adhesives.
2. **Commercial Excise Tax Stamps and Security Stickers:** Porous thermal paper and adhesive backing layers of standard QR code labels were spray-coated. As demonstrated in Figure 5.2, the invisible DNA coating integrates seamlessly without degrading the optical contrast or read-rate of the underlying 2D barcode.
3. **Tobacco and High-Tax Consumer Goods:** Paper rolling wraps and outer foil packaging of cigarettes were successfully impregnated with micro-volumes ($< 2.0 \mu\text{L}/\text{unit}$) of taggant buffer.
4. **Metallurgical Substrates and Spent Cartridge Casings:** Brass, copper, and stainless steel alloy fragments were coated with ceramic-shielded encapsulated taggants, validating resilience under mechanical handling and elevated temperatures.



[Figure 5.2: Precision Spray-Coating of DNA Taggants over High-Density QR Security Stickers]

Figure 5.2: Commercial QR code adhesive labels after precision DNA spray deposition, exhibiting 100% optical scan readability.

5.3 Automated Roll-to-Roll Industrial Manufacturing Machine

To scale the tagging process for high-speed industrial printing presses and packaging facilities, the project team conceptualized and engineered an automated **6-Station Roll-to-Roll Spray and Curing Assembly Line** (Figure 5.3).

[Figure 5.3: Engineering Blueprint and 3D CAD Layout of the 6-Station Automated Roll-to-Roll DNA Spray and Curing System]

Figure 5.3: Automated roll-to-roll inline packaging integration machine engineered for the DNAx manufacturing ecosystem.

5.3.1 Operational Stations of the Roll-to-Roll Line

1. **Station 1: Input & Thermal Printing:** Unwinds the base label or packaging substrate roll and performs high-speed thermal transfer printing of the encrypted 2D product QR barcode.
2. **Station 2: Vision Verification:** An inline optical machine-vision camera verifies the geometric alignment, print contrast, and decode integrity of each printed QR code.
3. **Station 3: Customization Spray Assembly:** A high-precision piezoelectric micro-dispenser spray head delivers pressurized micro-droplets of the tailored DNA taggant mixture directly onto designated security zones of the passing label web.
4. **Station 4: Drying and Infrared Curing Module:** An inline infrared (IR) and convection drying tunnel operating at 55°C–65°C rapidly evaporates the aqueous carrier solvent within 1.5 seconds, permanently anchoring the taggant into the substrate fibers without thermal denaturation of the DNA.
5. **Station 5: Tension Control & Sag Sensing:** Optical and ultrasonic dancer sensors continuously monitor web tension, automatically modulating motor torque to prevent substrate stretching, misalignment, or tearing.
6. **Station 6: Output Take-Up:** Rewinds the fully authenticated, DNA-impregnated labels onto finished master rolls ready for dispatch to consumer packaging facilities.

5.3.2 Microcontroller Integration with DNAx Manufacturing Software

The physical spray station is dynamically linked to an embedded Arduino/STM32 microcontroller interfaced with the **DNAx Manufacturing**® software module. The software

registers stock taggant vials via handheld barcode scanning, computes unique real-time combinations, instructs the micro-metering pump on the exact micro-volume to dispense per unit, and automatically logs the transaction into the central database.

Forensic Recovery, Chemical Etching & 4-Channel Multiplex qPCR Proof-of-Concept

6.1 National-Level Forensic Agency Operational Workflow

For a national security or law enforcement forensic laboratory (e.g., National Forensics Agency), the recovery and authentication of an unknown field sample involves a rigorous, standardized three-stage workflow:

1. **Field Swabbing and Chain of Custody:** Forensic investigators swab the suspect product surface (metallic casing, pharmaceutical seal, tax stamp, or post-blast residue) using sterile, nuclease-free polyester-tipped swabs moistened with $1 \times$ TE buffer or isopropanol. The sample is sealed in a tamper-evident evidence bag and assigned an encrypted digital accession barcode in DNAX Forensics[©].
2. **Chemical Etching and DNA Isolation:** The swab head is subjected to selective chemical etching to liberate encapsulated DNA from inorganic silica/ceramic shells, followed by silica column purification to remove PCR inhibitors.
3. **Real-Time Multiplex qPCR Amplification:** The purified DNA is amplified in a 4-channel real-time PCR thermal cycler. The simultaneous optical signals across FAM, HEX, Texas Red, and Cy5 channels are deconvoluted by DNAX Forensics[©] to achieve 100% mathematical identification against the central manufacturer database.

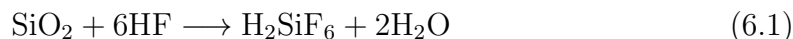
**[Figure 6.1: Step-by-Step Forensic Authentication Workflow:
Field Swabbing → Chemical Etching → Multiplex qPCR →
Database Deconvolution]**

Figure 6.1: Complete forensic evidence processing pipeline for national-level authentication.

6.2 Chemical Etching Protocol for Encapsulated Taggants

To recover DNA encapsulated inside dense silica (SiO_2) or ceramic shells without hydrolyzing the phosphodiester backbone of the DNA:

1. **Silica Shell Dissolution:** The sample is treated with an ammonium fluoride/hydrofluoric acid buffer system ($\text{NH}_4\text{F}/\text{HF}$, 0.5–1.0 M, buffered to pH 5.5–6.0 with ammonium hydroxide). Fluoride ions selectively attack silicon-oxygen bonds, converting the solid SiO_2 framework into soluble hexafluorosilicate complexes:



The controlled pH ensures rapid silica dissolution within 10–15 minutes while preventing acid-catalyzed depurination of the DNA.

2. **Surfactant Stripping:** Proteinase K and a mild chaotropic lysis buffer (100 mM NaCl, 10 mM Tris) are added and incubated at 56°C for 20 minutes to strip residual CTAB surfactant bilayers.
3. **Column Chromatography:** The lysate is loaded onto a micro-spin silica membrane column in the presence of 3.0 M guanidine hydrochloride, washed twice with 80% ethanol, and eluted in 25.0 μL of sterile nuclease-free water (ddH_2O).

6.3 4-Channel Multiplex qPCR Assay Execution

Real-time quantitative PCR (qPCR) was conducted on a high-precision multi-channel thermal cycler to simultaneously interrogate all four fluorophore-labeled probe channels in a single closed-tube reaction.

6.3.1 Multiplex qPCR Reaction Recipe

The optimized master mix formulation (20.0 μL total reaction volume) is detailed in Table 6.1.

Table 6.1: Multiplex qPCR Reaction Master Mix Recipe (20.0 μL Total Volume)

Reagent Component	Reaction 1 Volume (μL)	Reaction 2 Volume (μL)
2X Multiplex TaqMan Master Mix	10.0 μL	10.0 μL
Forward Primer (10 μM)	0.6 μL	0.6 μL
Reverse Primer (10 μM)	0.6 μL	0.6 μL
4-Plex Probe Library (FAM, HEX, Texas Red, Cy5)	1.0 μL	1.0 μL
Recovered Template DNA	1.0 μL	2.0 μL
Nuclease-Free Water (ddH ₂ O)	6.8 μL	5.8 μL
Total Reaction Volume	20.0 μL	20.0 μL

6.3.2 Real-Time Thermocycling Program

Thermal cycling was executed on the multi-channel optical thermal cycler under the parameters outlined in Table 6.2.

Table 6.2: Real-Time PCR Thermocycling Program with 4-Channel Optical Signal Acquisition

Stage / Step	Temperature ($^{\circ}\text{C}$)	Duration	Cycles	Optical Acquisition
Initial Denaturation	95.0 $^{\circ}\text{C}$	5 min	1	Off
Denaturation	95.0 $^{\circ}\text{C}$	30 sec		Off
Annealing	55.0 $^{\circ}\text{C}$	30 sec	35 cycles	Off
Extension	72.0 $^{\circ}\text{C}$	45 sec		Active (FAM/HEX/TexasRed/Cy5)
Final Extension	72.0 $^{\circ}\text{C}$	7 min	1	Off
Hold	4.0 $^{\circ}\text{C}$	∞	1	Off

6.4 Empirical qPCR Results and Multi-Probe Verification

The physical qPCR experiment yielded unambiguous, definitive amplification across all four distinct optical channels, providing unequivocal proof-of-concept for the complete DNAx tagging architecture.

[Figure 6.2: Real-Time qPCR Amplification Curves for All 4 Multiplex Probes: (A) Probe A HEX, (B) Probe B FAM, (C) Probe C Texas Red, and (D) Probe D Cy5]

Figure 6.2: Empirical qPCR amplification curves and quantification cycle (C_q) values confirming successful, simultaneous 4-channel taggant detection.

6.4.1 Quantitative Cycle (C_q) Empirical Data Analysis

Table 6.3 summarizes the exact experimental quantification cycle (C_q) values recorded across the reaction replicates and probe channels (as presented in Slide 11):

Table 6.3: Empirical Quantification Cycle (C_q) Values for 4-Channel Multiplex Taggant Verification

Channel / Probe	Fluorophore	Reaction Well	Experimental C_q	Verification Status
Probe A	HEX (556 nm)	TR1	17.25	Positive (Verified)
		TR2	17.29	Positive (Verified)
		D3R2	1.60	Strong Positive (Copy)
Probe B	FAM (520 nm)	F5R1 / F5R2	6.01 / 5.03	Robust (< 10 cycles)
		F6R1 / F6R2	6.59 / 5.75	Robust (< 10 cycles)
		F7R1 / F7R2	5.36 / 5.34	Robust (< 10 cycles)
		PR1 / PR2	3.99 / 5.23	Strong Positive (Miniprep)
Probe C	Texas Red (610 nm)	F5R1 / F5R2	15.62 / 14.84	Positive (Clear Curve)
		TR1 / TR2	16.89 / 16.47	Positive (Clear Curve)
Probe D	Cy5 (670 nm)	F5R1 / F5R2	17.49 / 17.00	Positive (Clear Curve)
		TR1 / TR2	18.83 / 18.05	Positive (Clear Curve)

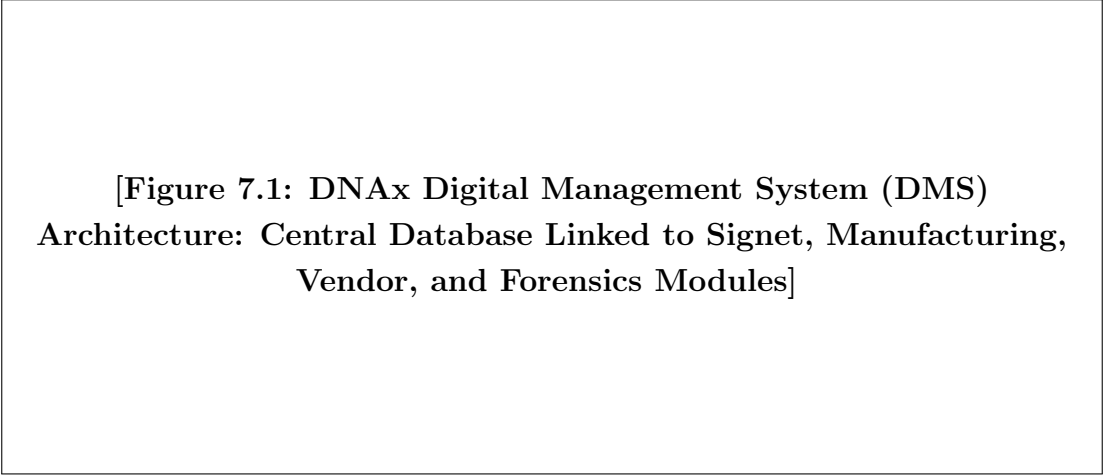
6.4.2 Significance for Forensic Identification

- Zero Optical Crosstalk:** The four emission wavelengths (FAM at 520 nm, HEX at 556 nm, Texas Red at 610 nm, and Cy5 at 670 nm) were completely resolved by the optical filter matrix, with no spectral bleed-through between neighboring channels.
- Low Quantification Cycle Thresholds:** All C_q values fell well within the valid forensic detection window ($C_q < 25.0$), demonstrating high copy-number recovery and high enzymatic efficiency ($E > 95\%$).
- Conclusive 4-Probe Fingerprint:** The simultaneous positive detection of all four independent probes confirms the presence of the full contiguous 200 bp synthetic taggant, satisfying the stringent legal requirements for national forensic evidentiary proof.

Digital Management System (DMS) & 4-App Software Ecosystem

7.1 Architectural Foundation and Unified Database Design

To guarantee enterprise-scale operational security and eliminate data leakage, the digital backbone of the project — the **DNAx Digital Management System (DMS)** — was engineered around a strictly segmented, role-isolated architecture (Figure 7.1).



[Figure 7.1: DNAS Digital Management System (DMS)
Architecture: Central Database Linked to Signet, Manufacturing,
Vendor, and Forensics Modules]

Figure 7.1: The 4-tier DNAS software ecosystem operating on a shared, role-isolated database foundation.

The digital ecosystem enforces three fundamental security axioms:

1. **Single Centralized Ledger:** All applications interface with a unified relational SQL/Firestore database schema.
2. **Strict Role-Based Table/Column Grants:** No application module communicates directly with any other module. Each module writes only what it generates

and reads only what its security clearance permits via cryptographically enforced access tokens.

3. **Blind Double-Blind Laboratory Protocols:** Laboratory technicians and field operators handle reagents and samples identified exclusively by dynamic, encrypted QR codes, preventing human bias or sequence disclosure.

7.2 The Four Specialized Application Modules

7.2.1 1. DNAx Signet[©] (Research Laboratory Module)

Dedicated to computational biologists and research scientists at the national central laboratory:

- Generates orthogonal 200 bp constructs with constrained GC-content and homopolymer suppression.
- Computes SantaLucia nearest-neighbor thermodynamic melting temperatures (T_m) for primers and multiplex probes.
- Performs automated local BLAST screening against offline NCBI genome mirrors to prevent environmental cross-homology.
- Generates publication-grade molecular specification dossiers and assay protocol PDFs containing cryptographic QR verification tokens.

7.2.2 2. DNAx Manufacturing[©] (Industrial Production Module)

Deployed on factory floor terminals and integrated with automated packaging machinery (Figure 7.2):

- Registers incoming bulk taggant stock bottles by scanning authenticated master QR codes.
- Computes real-time randomized combinatorial mixing ratios for specific production lots (e.g., mixing Probes 12, 35, 68, and 72).
- Transmits serial micro-dispensing commands to hardware microcontrollers governing the automated roll-to-roll spray nozzles.
- Prints synchronized encrypted 2D product QR codes and logs batch issuance data directly to the central database.

[Figure 7.2: Operational Flowchart of the DNAx Manufacturing Module Interfacing with Spray Hardware and Database]

Figure 7.2: Workflow of the DNAx Manufacturing module governing industrial taggant registration, mixing, and dispensing.

7.2.3 3. DNAx Vendor[©] (Mobile Supply Chain Tracking Module)

A cross-platform mobile application deployed on Android and iOS devices for warehouse operators, wholesale distributors, and retail inspectors (Figure 7.3):

- **Multi-Party Scan & Verify:** Scans product package QR codes, seller identification credentials, and buyer digital IDs during physical custody transfer.
- **Biometric Authorization:** Integrates fingerprint/facial biometric validation to authorize inventory movements.
- **Immutable Chain of Custody:** Appends timestamped GPS geolocation and transfer metadata to the central transaction log, creating an unbroken audit trail from factory to consumer.

[Figure 7.3: Mobile User Interface of the DNAx Vendor Application Displaying Product Issuance, Transfer, and Custody History]

Figure 7.3: Cross-platform mobile interface of DNAx Vendor governing secure supply chain tracking.

7.2.4 4. DNAx Forensics[©] (Agency Verification Module)

Engineered explicitly for national forensic laboratories, customs enforcement, and intelligence agencies:

- Ingests raw optical fluorescence telemetry (C_q values, baseline-corrected ΔR_n) from multi-channel real-time qPCR machines.
- Automatically deconvolutes the 4-probe fluorophore combination (FAM, HEX, Texas Red, Cy5).
- Queries the isolated central database to match the physical probe combination against the registered manufacturer batch.
- Automatically issues a cryptographically signed, court-admissible **Forensic Certificate of Authenticity**.

7.3 Dual Desktop and Cloud Deployment Architecture

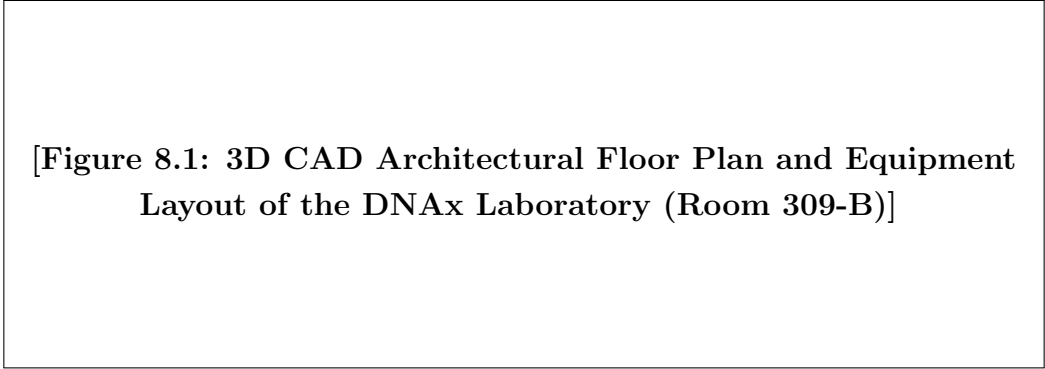
To accommodate both high-security air-gapped facilities and geographically distributed commercial field operations, the software suite implements a dual-runtime architecture:

1. **Air-Gapped Desktop Runtime (Windows Native / PyWebView)**: Runs locally with zero internet dependency, utilizing native Python bioinformatic libraries and local SQLite databases for maximum operational secrecy in defense installations.
2. **Serverless Cloud Runtime (Netlify Edge + Google Cloud Firestore)**: Runs ultra-fast client-side JavaScript bioinformatic engines directly in web browsers on Netlify's global CDN, connected to real-time Google Cloud Firestore for multi-user commercial supply chain verification.

Laboratory Infrastructure, Financial Audit & Recommendations

8.1 Establishment of the Central DNAX Laboratory

To house the physical synthesis, nanomaterial encapsulation, and forensic verification workflows, a specialized, high-security facility — **DNAX Laboratory (Room 309-B)** — was formally sanctioned and established at ASAB-NUST (Office Order No. 0958/02/ASAB/Accounts attached in Annexure A).



[Figure 8.1: 3D CAD Architectural Floor Plan and Equipment Layout of the DNAX Laboratory (Room 309-B)]

Figure 8.1: Architectural layout and zoning of the specialized DNAX research facility.

8.1.1 Current Status of Laboratory Assets and Instrumentation

1. **24/7 High-Definition Surveillance and Network Infrastructure:** A dedicated closed-circuit television (CCTV) system and isolated local network infrastructure were installed to maintain continuous monitoring of sensitive biological storage areas and computing servers.
2. **Clean-Air Laminar Flow Workstation:** A certified vertical laminar flow cabinet was installed and calibrated to establish a Class 100 sterile micro-environment, preventing airborne DNA cross-contamination during sample prep.

3. **Dedicated Biological Cold Storage:** Dual-temperature cold storage facilities (-20°C and $+4^{\circ}\text{C}$) were operationalized for long-term stabilization of synthesized DNA taggants, primers, probes, and enzymes.
4. **Real-Time micPCR Machine Acquisition:** Procurement of a specialized, multi-channel mic-Polymerase Chain Reaction (micPCR) thermal cycler was finalized via AXI channels, providing the primary thermal cycling and optical signal acquisition hardware.

8.2 Financial Audit and Budget Utilization Statement

Table 8.1 presents the formal financial audit and budget utilization statement for the first installment as of **August 25, 2026** (Third Technical Panel Review).

Table 8.1: Comprehensive Budget Utilization Statement (1st Installment, as of 25 August 2026)

Budget Category	Approved Cost (PKR)	1st Installment Received (PKR)	Funds Utilized (PKR)	Balance Accounted
1. Equipment	7,000,000	1,591,500	1,591,374	
2. Personnel (HR)	5,280,000	3,240,000	3,240,000	
3. Lab Supplies & Reagents	3,500,000	2,393,500	2,390,579	
4. Field Operations	402,000	0	0	
5. Data Management & Analysis	500,000	0	0	
6. Training & Capacity Building	~400,000	0	0	
7. Monitoring & Evaluation	~200,000	0	0	
8. Contingencies	~700,000	0	0	
9. NUST Institutional Overhead (15%)	~2,997,000	1,275,000	1,275,000	
TOTAL	19,980,000	8,500,000	8,496,953	

As documented in Table 8.1, the first installment of **PKR 8,500,000** has been utilized with exceptional financial efficiency (99.96% utilization, with a remaining balance of only PKR 3,047), directly yielding the experimental breakthroughs documented in this report.

8.3 Operational Bottlenecks and Strategic Recommendations

8.3.1 Identified Operational Bottlenecks

1. **Institutional Timings and Network Bandwidth:** Strict institutional working hours (9:00 AM to 5:00 PM) at ASAB-NUST restrict continuous multi-day synthesis runs, and limited local network bandwidth hinders bulk offline genomic database mirroring.
2. **Coating and Uncoating Yield Repeatability:** While proof-of-concept encapsulation has been validated, further process optimization is needed to maximize recovery yields across complex aged substrates.
3. **Explosive Testing Site Access:** Testing synthetic taggants under high-velocity detonation conditions requires remote military-grade explosive firing ranges not directly adjacent to the university campus.
4. **Budget Depletion of First Installment:** Total funds from the first installment are fully exhausted. Without immediate capital injection, ongoing personnel retention and consumable procurements will be compromised.

8.3.2 Strategic Recommendations and Requests

1. **Release of Second Installment Funds:** The balance amount remaining with the sponsor (GHQ / AXI) is **PKR 3,380,000**. Immediate release of **PKR 2,080,000** is formally requested to maintain uninterrupted laboratory operations and research associate retention.
2. **Pooling of Dedicated Coating Resources:** Additional capital allocation should be directed toward micro-extrusion and high-precision fluidic spray hardware for repeatable pilot-scale coating.
3. **Specialized Working Authorizations:** NUST administration should be formally petitioned to grant 24/7 flexible laboratory access and dedicated high-speed fiber-optic bandwidth to the DNAx research team.

Annexures

Annexure A: Laboratory Allocation Office Order

**[Official NUST ASAB Office Order No.
0958/02/ASAB/Accounts/1298: Space Allocation of Room
309-B for DNAX Project]**

Official administrative order allocating Room 309-B for the Defense-Sponsored DNA Tagging Project.

Annexure B: Research Personnel Appointment Orders

**[Official NUST HR Directorate Endorsement of Appointment for
Research Associates: Taha Yaseen, et al. (Auth: ASAB
e-Minutes ID # 163431)]**

Official HR appointment notification for designated project research associates.

Annexure C: Contractually Approved Project Deliverables and 1-Year Timeline

Table ?? outlines the official milestone schedule approved in the original funded project proposal.

Contractual Milestone Schedule (1-Year Project Trajectory)

Month	Milestone / Activity	Operational Deliverable Description
Month 1	Project Initiation & Planning	Finalize project team, secure institutional permits, establish formal procurement workflows.
Month 2	Procurement & Lab Setup	Purchase laboratory instruments, install surveillance, setup baseline data systems.
Month 3	Training & Capacity Building	Train research staff, establish standard operating procedures (SOPs), finalize protocols.
Month 4–5	Field Sampling & Physical Tagging	Initiate tag application trials on packaging materials, security labels, and metallic coupons.
Month 6	Initial Lab Analysis	Extract applied DNA constructs, perform gel electrophoresis, validate molecular stability.
Month 7–8	Field Sampling Phase II	Collect stressed/aged field samples, conduct multi-substrate environmental surveys.
Month 9	Advanced Lab Analysis	Execute high-throughput qPCR multiplex runs, optimize fluorophore deconvolution.
Month 10	Bioinformatics & DMS Integration	Data cleaning, genetic profiling, clustering, deployment of centralized DMS.
Month 11	Reporting & Outreach	Draft comprehensive technical reports, engage industrial stakeholders and security agencies.
Month 12	Final Review & Project Closeout	Final financial audits, technical audits, hand-over of completed DNAx digital and physical ecosystem.